

# EXPERIMENT 3

Determination and plotting of Force of attraction between two electric Charges and two parallel conductors separated by varying distance.

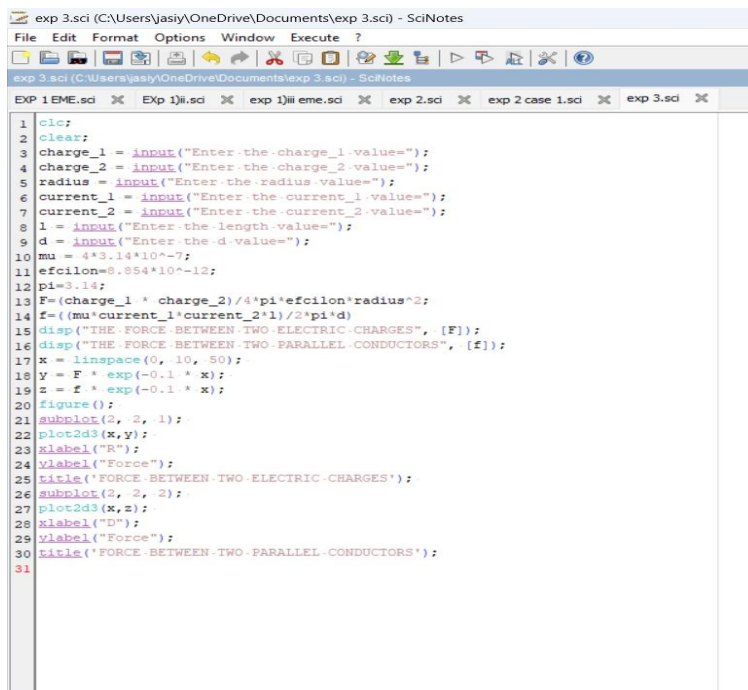
## Aim:

To Determination and plotting of Force of attraction between two electric Charges and two parallel conductors separated by varying distance between them using SCILAB.

## Software required:

- SCILAB version 6.0.1

## Program:



```
exp 3.sci (C:\Users\jasy\OneDrive\Documents\exp 3.sci) - SciNotes
File Edit Format Options Window Execute ?
exp 3.sci (C:\Users\jasy\OneDrive\Documents\exp 3.sci) - SciNotes
EXP 1 EME.sci EXP 1 IJ.sci exp 1 IJ EME.sci exp 2.sci exp 2 case 1.sci exp 3.sci
1 clc;
2 clear;
3 charge_1 = input("Enter the charge_1 value=");
4 charge_2 = input("Enter the charge_2 value=");
5 radius = input("Enter the radius value=");
6 current_1 = input("Enter the current_1 value=");
7 current_2 = input("Enter the current_2 value=");
8 l = input("Enter the length value=");
9 d = input("Enter the d value=");
10 mu = 4*3.14*10^-7;
11 efcilon=8.854*10^-12;
12 pi=3.14;
13 F=(charge_1 * charge_2)/4*pi*efcilon*radius^2;
14 f=(mu*current_1*current_2*l)/2*pi*d;
15 disp("THE FORCE BETWEEN TWO ELECTRIC CHARGES", [F]);
16 disp("THE FORCE BETWEEN TWO PARALLEL CONDUCTORS", [f]);
17 x = linspace(0, 10, 50);
18 y = F * exp(-0.1 * x);
19 z = f * exp(-0.1 * x);
20 figure();
21 subplot(2, 2, 1);
22 plot2d3(x,y);
23 xlabel("R");
24 ylabel("Force");
25 title('FORCE BETWEEN TWO ELECTRIC CHARGES');
26 subplot(2, 2, 2);
27 plot2d3(x,z);
28 xlabel("D");
29 ylabel("Force");
30 title('FORCE BETWEEN TWO PARALLEL CONDUCTORS');
31
```

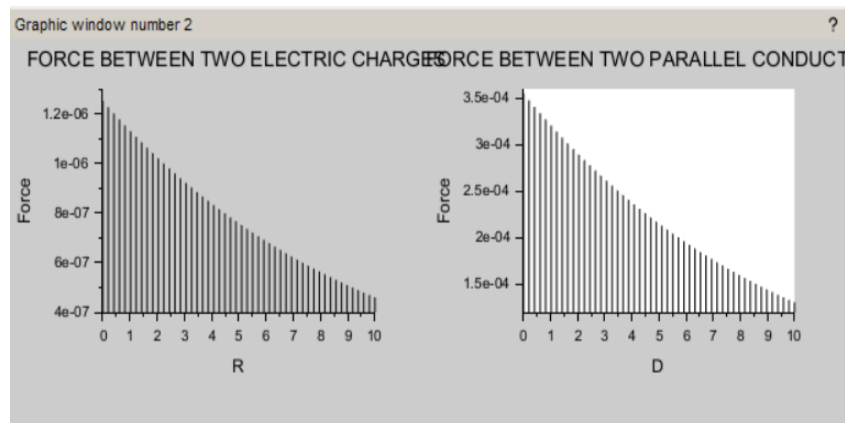
## Output:

```

Enter the charge_1 value=1
Enter the charge_2 value=2
Enter the radius value=3*10^2
Enter the current_1 value=3
Enter the current_2 value=4
Enter the length value=5
Enter the d value=3

"THE FORCE BETWEEN TWO ELECTRIC CHARGES"
0.0000013
"THE FORCE BETWEEN TWO PARALLEL CONDUCTORS"
0.0003549
-1-> |

```



## Theoretical calculation:

EXP No: 03

Determination and plotting of Force of attraction between two electric charges and two parallel conductors separated by varying distance.

Output:

Enter the charge -1 value = 1

Enter the charge -2 value = 2

Enter the charge -3 value = 3

Enter the radius value =  $3 \times 10^{-2}$

Enter the current -1 value = 3

Enter the current -2 value = 4

Enter the length value = 5

Enter the d value = 3

1. Force between two electric charges

Coulomb's law  $F = \frac{1}{4\pi\epsilon_0} \times \frac{q_1 q_2}{r^2}$

$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$

$F = \frac{9 \times 10^9 \times 1 \times 2}{(3 \times 10^{-2})^2}$

$F = \frac{18 \times 10^9}{90000}$

$F = 2 \times 10^{-9} \text{ N}$

2. Force between two parallel conductors

$F = \frac{\mu_0 I_1 I_2}{2\pi d} \times L$   $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$

$F = \frac{4\pi \times 10^{-7} \times 3 \times 4 \times 5}{2\pi \times 3} = \frac{4 \times 10^{-7} \times 60}{6} = \frac{160 \times 10^{-7}}{6} = 2.67 \times 10^{-5}$

$F = \frac{4\pi \times 10^{-7} \times 3 \times 4 \times 5}{2\pi \times 3} = \frac{240 \times 10^{-7}}{6\pi} = 4 \times 10^{-7} = 4.0 \times 10^{-6} \text{ N}$

PRACTICAL OUTPUT :

"The force between two electric charges"

0.0000013

"The force between two parallel conductors"

0.0003549

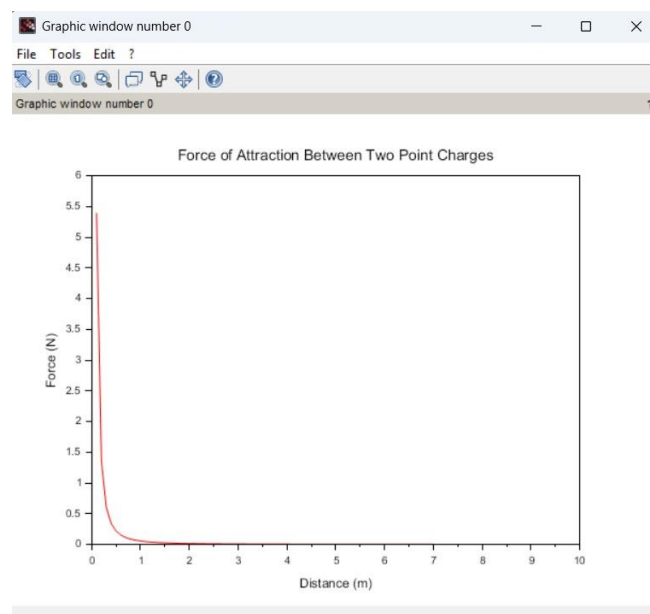
## Case 1 :

Given two point charges  $q_1=2 \times 10^{-6}$  and  $q_2=3 \times 10^{-6}$  C, calculate and plot the force of attraction between them as a function of distance, ranging from 0.1 m to 10 m. Explain how the force changes with the distance between the charges.

## Code:

```
exp 3case.lsci (C:\Users\yasi\OneDrive\Documents\exp 3case.lsci) - SciNotes
File Edit Format Options Window Execute ?
exp 3case.lsci (C:\Users\yasi\OneDrive\Documents\exp 3case.lsci) - SciNotes
EXP 1 EME.sci exp 1 il.sci exp 1 il eme.sci exp 2.sci exp 2 case 1.sci exp 3.sci exp
1 // Parameters for two point charges
2 ke = 8.99e9; // Coulomb's constant (N*m^2/C^2)
3 q1 = 2e-6; // Charge 1 (Coulombs)
4 q2 = 3e-6; // Charge 2 (Coulombs)
5 r = linspace(0.1, 10, 100); // Distance between charges (m)
6
7 // Calculate force as a function of distance
8 F = ke * abs(q1 * q2) ./ r.^2;
9
10 // Plotting the force vs distance
11 plot(r, F, "r");
12 xlabel("Distance (m)");
13 ylabel("Force (N)"); // Parameters for two point charges
14 ke = 8.99e9; // Coulomb's constant (N*m^2/C^2)
15 q1 = 2e-6; // Charge 1 (Coulombs)
16 q2 = 3e-6; // Charge 2 (Coulombs)
17 r = linspace(0.1, 10, 100); // Distance between charges (m)
18
19 // Calculate force as a function of distance
20 F = ke * abs(q1 * q2) ./ r.^2;
21
22 // Plotting the force vs distance
23 plot(r, F, "r");
24 xlabel("Distance (m)");
25 ylabel("Force (N)");
26 title("Force of Attraction Between Two Point Charges");
27
```

## Output:



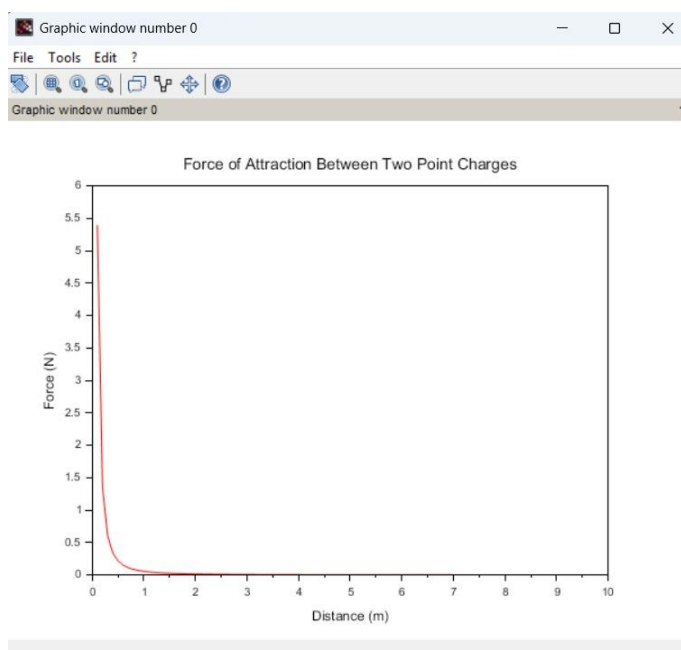
## Case 2:

Given two point charges  $q_1 = 5 \times 10^{-6} \text{ C}$  and  $q_2 = 2 \times 10^{-6} \text{ C}$ , calculate and plot the force of attraction between them as the distance varies from 0.5 m to 5 m. Discuss how the force behaves with the varying distance and charge magnitudes.

## Code:

```
EXP 1 EME.sci  EXP 1 IJ.sci  EXP 1 IJ EME.sci  EXP 2.sci  EXP 2 case 1.sci  EXP 3.sci  EXP 3 case  
1 // Parameters for two point charges  
2 clc;  
3 ke = 8.99e9; // Coulomb's constant (N*m^2/C^2)  
4 q1 = 5e-6; // Charge 1 (Coulombs)  
5 q2 = 2e-6; // Charge 2 (Coulombs)  
6 r = linspace(0.5, 5, 100); // Distance between charges (m)  
7  
8 // Calculate force as a function of distance  
9 F = ke * abs(q1 * q2) ./ r.^2;  
10  
11 // Plotting the force vs distance  
12 plot(r, F, "b");  
13 xlabel("Distance (m)");  
14 ylabel("Force (N)");  
15 title("Force of Attraction Between Two Point Charges (Different Magnitudes)");  
16  
17
```

## Output:



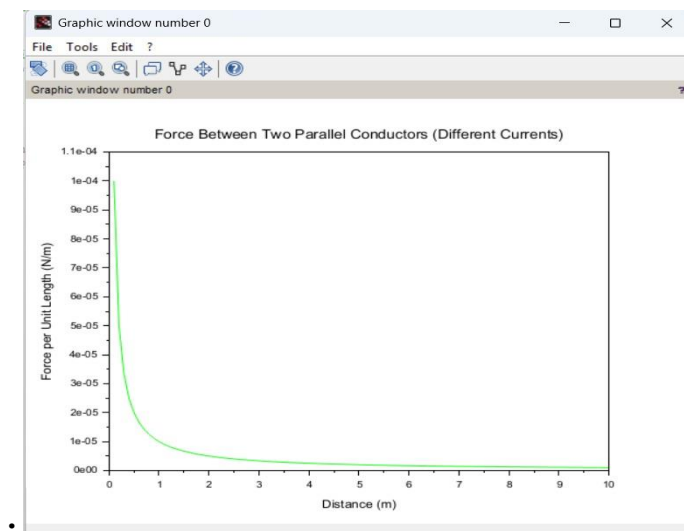
### Case 3:

Two parallel conductors are separated by a distance  $d$ . The first conductor carries a current of  $I_1=5$  A, and the second conductor carries a current of  $I_2=10$  A. Calculate and plot the force per unit length between the conductors for varying distances from 0.1 m to 10 m. Explain how the force changes with the distance between the conductors.

### Code:

```
exp 3 case iv.sci (C:\Users\jasiy\OneDrive\Documents\exp 3 case iv.sci) - SciNotes
File Edit Format Options Window Execute ?
exp 3 case iv.sci (C:\Users\jasiy\OneDrive\Documents\exp 3 case iv.sci) - SciNotes
EXP 1 EME.sci EXP 1 IIS.sci exp 1 IIS eme.sci exp 2.sci exp 2 case 1.sci exp 3.sci
1 // Parameters for two parallel conductors
2 clc;
3 mu0 = 4 * pi * 1e-7; // Permeability of free space (N/A^2)
4 I1 = 5; // Current in conductor 1 (Amperes)
5 I2 = 10; // Current in conductor 2 (Amperes)
6 d = linspace(0.1, 10, 100); // Distance between conductors (m)
7
8 // Calculate force per unit length as a function of distance
9 F_per_length = (mu0 * I1 * I2) ./ (2 * pi * d);
10
11 // Plotting the force per unit length vs distance
12 plot(d, F_per_length, "m");
13 xlabel("Distance (m)");
14 ylabel("Force per Unit Length (N/m)");
15 title("Force Between Two Parallel Conductors (Same Current)");
16
```

### Output:



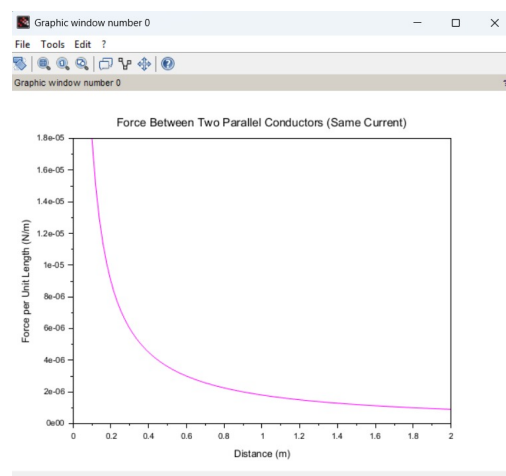
## Case 4:

Given two parallel conductors carrying currents of 3 A, calculate and plot the force per unit length between the conductors for varying distances between 0.1 m to 2 m. Discuss the relationship between the distance and the force between the conductors.

## Code:

```
exp 3 case iv.sci (C:\Users\yasi\OneDrive\Documents\exp 3 case iv.sci) - SciNotes
File Edit Format Options Window Execute ?
exp 3 case iv.sci (C:\Users\yasi\OneDrive\Documents\exp 3 case iv.sci) - SciNotes
EXP 1 EME.sci EXP 1 j.sci exp 1 j eme.sci exp 2.sci exp 2 case 1.sci exp 3.sci
1 // Parameters for two parallel conductors
2 clc;
3 mu0 = 4 * %pi * 1e-7; // Permeability of free space (N/A^2)
4 I1 = 3; // Current in conductor 1 (Amperes)
5 I2 = 3; // Current in conductor 2 (Amperes)
6 d = linspace(0.1, 2, 100); // Distance between conductors (m)
7
8 // Calculate force per unit length as a function of distance
9 F_per_length = (mu0 * I1 * I2) ./ (2 * %pi * d);
10
11 // Plotting the force per unit length vs distance
12 plot(d, F_per_length, 'm');
13 xlabel('Distance (m)');
14 ylabel('Force per Unit Length (N/m)');
15 title('Force Between Two Parallel Conductors (Same Current)');
16
```

## Output:



**Result :**

Thus the program was verified for the force of attraction or repulsion acting along a straight line between two electric charges is directly proportional to the product of the charges and inversely to the square of the distance between them using SCILAB was successfully.